Pre-Lie algebras and F-manifolds

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(Nonassociative) algebras [over an infinite field of characteristic zero] for a given signature $\mathcal{S} \rightsquigarrow$ an *operad* $\mathcal{O}_{\mathcal{S}}$.

Operads are *associative* algebras in a certain monoidal category, and algebras for the signature $\mathcal{S}$ are examples of modules over $\mathcal{O}_{\mathcal{S}}$.

Some results on nonassociative algebraic structures become easier when attempted *syntactically* using the interpretation of algebraic structures as associative algebras.

Thus, nonassociative mathematics occasionally reduces to associative mathematics (in a much bigger monoidal category).

Motivation

Frobenius manifolds : an important class of Riemannian manifolds which arise in algebraic geometry, integrable systems, symplectic geometry, etc.

A Frobenius manifold is a Riemannian manifold whose tangent sheaf is given a structure of a sheaf of Frobenius algebras (commutative associative algebras with $(X \cdot Y, Z) = (X, Y \cdot Z)$), with an extra *potentiality* condition for the structure constants (also known as the WDVV equation).

Potentiality: the expression $(X \cdot Y, Z)$ is of the form $\partial_X \partial_Y \partial_Z(H)$ for some function H, the potential associated to the given Frobenius manifold.

Motivation

Hertling and Manin '98 : suppose that M is a (not necessarily Riemannian) manifold for which the tangent sheaf $\mathcal{T}_M$ is given a structure of a sheaf of commutative associative algebras. The expression

$$\begin{aligned} HM &:= [X \cdot Y, Z \cdot T] \\ &- [X \cdot Y, Z] \cdot T - [X \cdot Y, T] \cdot Z - [X, Z \cdot T] \cdot Y - [Y, Z \cdot T] \cdot X \\ &+ [X, Z] \cdot Y \cdot T + [X, T] \cdot Y \cdot Z + [Y, Z] \cdot X \cdot T + [Y, T] \cdot X \cdot Z \end{aligned}$$

is a tensor (is $\mathcal{O}_M$-multilinear).

For a Frobenius manifold, this tensor vanishes.

Motivation

A manifold for which the tensor HM identically vanishes is called an F-manifold (or a weak Frobenius manifold).

A good way to remember the definition of the Hertling–Manin tensor: introduce the expression

$$P_X(Y,Z) = [X, Y \cdot Z] - [X,Y] \cdot Z - [X,Z] \cdot Y,$$

the obstruction for $\mathcal{T}_M$ to be a sheaf of Poisson algebras.

Then we have

$$HM = P_{X \cdot Y}(Z,T) - P_X(Z,T) \cdot Y - P_Y(Z,T) \cdot X.$$

Thus, the F-manifold condition can be viewed as a relaxed version of the Poisson condition.

Motivation

In the case of Poisson algebras, there is a well known result which relates them to (noncommutative) associative algebras. Physicists would say that Poisson algebras are "quasi-classical limits of nearly commutative" associative algebras".

In other words, if $a \star b$ is an $k[[\hbar]]$-bilinear associative product on $A[[\hbar]]$, and the induced product on $A = A[[\hbar]]/(\hbar)$ is commutative, then there is a canonically defined bracket on A defined as the operation on $A[[\hbar]]/(\hbar)$ induced by the operation $(a \star b - b \star a)/\hbar$ on $A[[\hbar]]$. The vector space A with this bracket and its commutative associative product becomes a Poisson algebra.

In terms of *operads*, the operad **Assoc** has a suboperad isomorphic to **Lie**. We may consider the filtration of the operad **Assoc** by powers of the operadic ideal generated by **Lie**. The associated graded operad is the operad **Poisson**.

Motivation

It is quite natural to consider the following "equation":

$$\frac{???}{\textbf{FMan}} = \frac{\textbf{Assoc}}{\textbf{Poisson}},$$

where **FMan** is the operad encoding "F-manifold algebras", that is vector spaces equipped with a commutative associative product $a, b \mapsto a \cdot b$ and a Lie bracket $a, b \mapsto [a, b]$. Similarly to the Poisson case, we move to a purely algebraic context, so we do not have sheaves of algebras, but really algebraic structures on a single vector space.

Here and below I shall use the language of operads, considering algebras of a given type syntactically rather than semantically. Informally, an operad encoding a given algebraic structure $\mathcal{S}$ is the collection of vector spaces of multilinear operations one can define using the syntax of $\mathcal{S}$.

The operad of F-manifold algebras

For instance, the space **FMan**(2) is the span of the structure operations, and the space **FMan**(3) is the span of the operations sending a, b, c to

$$\begin{gathered} a \cdot b \cdot c, [a, [b, c]], [b, [a, c]], \\ a \cdot [b, c], b \cdot [a, c], c \cdot [a, b], \\ [a, b \cdot c], [b, a \cdot c], [c, a \cdot b]. \end{gathered}$$

The space **FMan**(4) is 64-dimensional.

The sequence $2, 9, 64, \ldots$ is easily identifiable: according to Cayley, it counts labelled rooted trees. In the context of operads, it famously arises when working with pre-Lie algebras.

Pre-Lie algebras

Let us recall the notion of a pre-Lie algebra. (A Koszul–Vinberg algebra, a right-symmetric algebra, etc.)

A vector space A with a binary operation $\circ$ is called a (right) pre-Lie algebra if the associator of $\circ$ is right-symmetric:

$$(a \circ b) \circ c - a \circ (b \circ c) = (a \circ c) \circ b - a \circ (c \circ b).$$

Many examples:

- (Cayley–)Koszul–Vinberg, a manifold with a flat torsion-free connection, $X \circ Y = \nabla_Y(X)$,
- Gerstenhaber: structures on the Hochschild complex of an associative algebra,
- Agrachev–Gamkrelidze: "chronological algebras" needed to study non-stationary vector fields,
- Connes–Kreimer: appearance in the mathematical formalism of renormalisation,
- Chapoton–Livernet(–Méndez): rooted trees with labelled vertices and free pre-Lie algebras.

Pre-Lie algebras, Lie algebras etc.

The main reason for a (slightly misleading) term "pre-Lie" is that in any pre-Lie algebra the operation $[a, b] = a \circ b - b \circ a$ is a Lie bracket, that is satisfies the Jacobi identity. In terms of operads, the operad **PreLie** has a suboperad isomorphic to **Lie**.

Suppose that a pre-Lie algebra is commutative. Then the structure identity becomes

$$(a \circ b) \circ c = (a \circ c) \circ b,$$

which due to commutativity is the same as

$$(b \circ a) \circ c = b \circ (a \circ c),$$

so a commutative pre-Lie algebra is automatically associative. In terms of operads, the quotient of the operad **PreLie** by the ideal generated by **Lie** is the operad **Com** of commutative associative algebras.

Main theorem

Consider the filtration of the operad **PreLie** by powers of the operadic ideal generated by **Lie**. The associated graded operad is the operad **FMan** encoding F-manifold algebras.

Outline of the proof

Semantic part: relations of **FMan** hold in $\mathrm{gr}(\mathbf{PreLie})$. (This is a direct computation.) Thus, $\dim(\mathbf{FMan}(n)) \geq \dim(\mathbf{PreLie}(n))$, so we have a lower bound.

Outline of the computation: first, polarise, putting $a \cdot b = a \circ b + b \circ a$, $[a, b] = a \circ b - b \circ a$:

$$(a \cdot b) \cdot c - a \cdot (b \cdot c) - a \cdot [b, c] - [a, b] \cdot c - 2[a, c] \cdot b + \\ [a, b \cdot c] + [a \cdot b, c] + [[a, c], b] = 0,$$

Then do what people do when computing Gröbner bases: rewrite $((ab)c)d$ in two different ways using this relation. The result, once we pass to $\mathrm{gr}(\mathbf{PreLie})$, becomes the HM relation.

Outline of the proof

Syntactic part, step 1: view relations of **FMan** as a rewriting system on the categorical coproduct of **Com** and **Lie**.

The rewriting rule is

$$[a \cdot b, c \cdot d] \to \textit{something},$$

where *something* is a combination of monomials where the first operation applied is the product.

It is possible to prove (in fact in much higher generality) that the rewriting rule

$$[a \cdot b, c \cdot d] \to 0$$

gives an operad whose dimensions of components are not less than those of **FMan** (upper bound). A rigorous proof uses the machinery of "shuffle operads" (D.–Khoroshkin).

Outline of the proof

Syntactic part, step 2: consider the quotient of the categorical coproduct of **Com** and **Lie** by the relation

$$[a \cdot b, c \cdot d] = 0.$$

The idea is to apply methods of homotopical algebra for operads, and construct a free resolution of this operad. Using the terminology and notation of Koszul duality for operads, the minimal model of this operad is generated by

$$s^{-1}\overline{\mathbf{Com}^{¡}} \oplus s^{-1}\overline{\mathbf{Lie}^{¡}} \oplus (\overline{\mathbf{Lie}^{¡}} \circ (s^{-1}\overline{\mathbf{Com}^{¡}})).$$

Roughly speaking, the first term provides a coherent system of homotopies for **Com**, the second one does it for **Lie**, and the third term deals with the way associativity and the Jacobi identity interact with the new relations. (The proof is not very hard, but it is overall very surprising that an explicit minimal resolution exists, since this operad is beyond the scope of usual methods of the operad theory.)

Outline of the proof

For any operad $\mathcal{P}$ and its minimal model $(\mathcal{T}(\mathcal{E}), d)$, there is a relationship between dimensions of components of $\mathcal{P}$ and the Euler characteristics of components of $\mathcal{E}$. In our case, we establish that dimensions of components of the quotient of the categorical coproduct of **Com** and **Lie** by the relation

$$[a \cdot b, c \cdot d] = 0.$$

are given by the Cayley sequence $\{n^{n-1}\}$.

Altogether, we now have an upper and a lower bound on the operad **FMan** which are the same, therefore the natural surjection

$$\mathbf{FMan} \twoheadrightarrow \mathrm{gr}(\mathbf{PreLie})$$

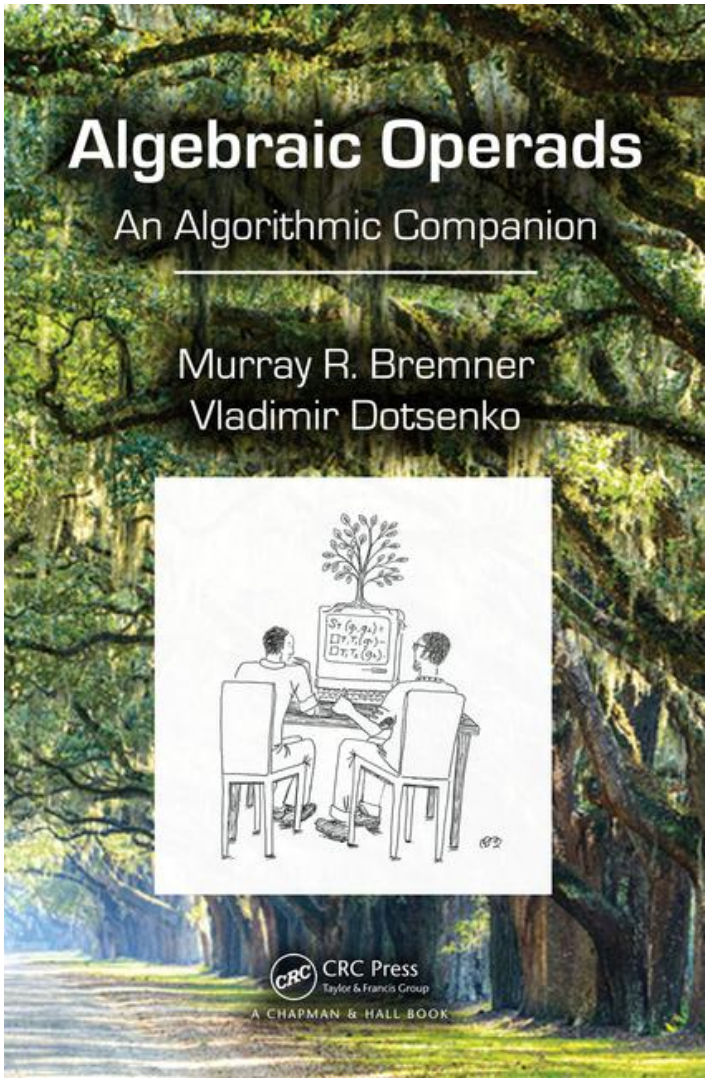
(semantics) is an isomorphism.

Concluding remarks

"Deformation quantisation" for F-manifolds? Unlike the Poisson case, there are obstructions. Those obstructions are quite meaningful, and make one consider algebraic structures related to the so called *flat F-manifolds*.

As a consequence of the fact that the lower bound is sharp, the rewriting system defining the operad **FMan** is convergent (terminating and confluent). As a consequence, the operad **FMan** admits a resolution of the same shape as discussed above, allowing to work efficiently with F-manifold algebras up to homotopy. This connects to geometric work of Merkulov on strong homotopy F-manifolds.

Homotopical algebra is useful, even in topics where there is no homology or homotopy on the surface!

Thank you for your attention!

For more information about operads: